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Genomic analyses reveal potential independent adaptation to high altitude in Tibetan chickens

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Abstract

Much like other indigenous domesticated animals, Tibetan chickens living at high altitudes (2,200~4,100 m) show specific physiological adaptations to the extreme environmental conditions of the Tibetan Plateau, but the genetic bases of these adaptations are not well characterized. Here, we assembled a de novo genome of a Tibetan chicken, and re-sequenced whole genomes of 32 additional chickens, including Tibetan chickens, village chickens, game fowl, and Red Junglefowl, and found that the Tibetan chickens could broadly be placed into two groups. Further analyses revealed that several candidate genes in the calcium-signaling pathway are possibly involved in adaptation to the hypoxia experienced by these chickens, as these genes appear to have experienced directional selection in the two Tibetan chicken populations, suggesting a potential genetic mechanism underlying high altitude adaptation in Tibetan chickens. The candidate selected genes identified in this study, and their variants, may be useful targets for clarifying our understanding of the domestication of chickens in Tibet, and might be useful in current breeding efforts to develop improved breeds for the highlands.

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Introduction

Domesticated animals have spread to every corner of the globe alongside their human domesticators. As both humans and their accompanying domesticates spread, both have encountered new and sometimes harsh environments. Some of these migrations ended disastrously, but in others, both humans and their accompanying animals successfully adapted to novel environments—one such example would be the Tibetan Plateau. Ranging from 2,200~4,100 m in elevation, the Tibetan Plateau is among the harshest environments settled by humans. Tibetan wild animals, Tibetans, and some of their domesticated animals have developed effective strategies through specific physiological and genetic adaptations to survive at high altitude, particularly to address the low-oxygen hypoxic environment (see reviews (Monge and Leon-Velarde 1991; Ramirez et al. 2007; Storz et al. 2010).

The Tibetan chicken, an aboriginal breed with a documented breeding history of more than a millennia at high altitude, has blood with an increased number of red blood cells and a higher hemoglobin concentration, with the hemoglobin having a stronger affinity for oxygen (Zhang et al. 2007; Wu and Li 2012), as well as several histological adaptations (Wu and Li 2012). The peculiarities of many of the Tibetan chicken's phenotypic traits, as well as its origin, make it an interesting case for examining selective evolutionary pressure. However, such examinations are complicated by a number of factors. First, domesticated chickens are phenotypically incredibility diverse; Darwin noted that chickens possessed the greatest diversity of

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any known bird (Darwin 1868). Similarly, the likely progenitor of the domestic chickens, the Red Junglefowl, whom has already had its genome sequenced (Hillier et al. 2004), also possess great diversity with several different identified subspecies and multiple lineages that have likely contributed to the breeding of domesticated chickens (Liu et al. 2006). Finally, while the domestication of the chicken is comparatively well characterized (Rubin et al. 2010), much of this process is still unknown. To further our understanding of both high-altitude adaptation and the evolutionary history of domesticated animals, such as the chicken, the generation of additional genomes (especially those generated by de novo assembly) from different breeds of domesticated chickens as well as different wild Red Junglefowls should be of great assistance. In particular, using a cross-section of different breeds from a distinct geographic region with both highland and lowland areas that are known to have been settled within a relatively short time by similar human populations, to help control for some of the intrinsic diversity of chickens, should aid in examining high altitude adaptation and domestication. Here, we assembled a de novo genome of a Tibetan chicken, and re-sequenced whole genomes from 32 chickens, from Tibet and lowland areas from neighboring regions, which included specimens of the Tibetan chicken, Yunnan native village chicken (raised for food), as well as a Xishuangbanna game fowl raised for cockfighting, and Red Junglefowl.

Results and Discussion

A de novo assembled Tibetan chicken genome

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Multiple origins for the domestic chicken has been previously been reported (Liu et al. 2006), a factor complicating the well-known genetic diversity of chickens. There are likely numerous variations (e.g. large structural variation) that re-sequencing genome data cannot identify. Here, a draft genome of a female domestic Tibetan chicken sampled from the Tibetan Plateau was sequenced via whole genome shortgun strategy on an Illumina sequencing platform, which should be helpful for studying the artificial selection and adaptation of Tibetan chickens. In total, 256.5 Gb of raw sequence data was assembled into a genome with 162-fold sequence coverage, with N50 for scaffolds of 2.02 Mb, obtained using SOAPdenovo (Li et al. 2010) (**Supplementary Table 1-4**) and 632 scaffolds of >~310kb that cover 90% of the total genome sequence with the largest scaffold spanning 10.97Mb. A total of 17,462 protein-coding and 773 non-coding RNA genes were annotated in this genome, following an integration of genomic and transcriptomic data (**Supplementary Table 5-8**).

To evaluate the quality of the genome assembly, scaffolds of the Tibetan chicken were mapped onto the Red Junglefowl reference genome sequence using whole genome sequence alignment. Roughly 98.0 % of the Red Junglefowl assembled chromosome sequences were covered by scaffolds from the Tibetan chicken (**Supplementary Table 9**). Genes orthologous to Tibetan chicken genes were identified in the Red Junglefowl, Turkey, Darwin finch, and Zebra finch. Among these orthologous genes, 38 orthologs of Tibetan chicken genes were found in at least two other bird genomes (i.e., Turkey, Darwin finch or Zebra finch), but not in the Red

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Junglefowl genome sequence. This observation suggests that these 38 genes were either deleted or were not sequenced in the current Red Junglefowl genome assembly (**Supplementary Table 10**). We believed that our de novo genome for the Tibetan chicken would be helpful for studying adaptation in domestic chickens, particularly the Tibetan chickens. For the purposes of this study, we focus on candidate elements that might potentially contribute to the genetic basis of adaptation among Tibetan chickens.

When the genomes of Tibetan animals were compared with their counterpart species that live at lower altitudes, it was reported that the size of their olfactory receptor gene family was reduced (Qu et al. 2013; Li et al. 2014), which was attributed possibly to adaptation to a highland environment where food sources and odorants are limited, odorants diffuse more slowly, and the interactions between odorants and receptors are weakened (Kuehn et al. 2008). De novo assembled genomes rather than re-sequenced genomes may better facilitate the identification the size of the gene repertoire underlying olfactory receptors. Here, many of the Red Junglefowl olfactory receptor genes were also not identified in our Tibetan chicken genome assembly (**Supplementary Table 12, Supplementary Fig. 7**). However, when sequence reads from the Tibetan chicken were mapped to the Red Junglefowl genome, coverage of lost *OR* genes was similar to the average coverage of the genome (**Supplementary Table 13**). We also noticed a marked reduction in the number of segmental duplications in the Tibetan chicken, Turkey, and Darwin finch relative to the Red Junglefowl genomes (**Supplementary Table 11**). It is hard to

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determine whether the loss of *OR* genes in the Tibetan chicken genome is a simply a failure of the WGS assembly technique to identify segmental duplications, or whether they are true losses attributable to high altitude adaptation. Assembly of another genome of Red Junglefowl genome using the same strategy used for the Tibetan chicken should clarify this issue. Previous reports using the WGS-strategy for the Tibetan wild boar (Li et al. 2014) and the ground tit (Qu et al. 2013) provide cautionary tales on jumping to conclusions regarding the role of adaptive evolution in the contraction of the *OR* gene family size.

Population variation analyses revealed two independent groups of Tibetan chickens

Given the genetic diversity among domesticated chickens in general and the Tibetan chicken specifically, it would be better to study a larger umber of individuals to understand population variations in the Tibetan chicken. To better characterize directional selection acting on the genome of the Tibetan chicken, we re-sequenced whole genomes from 5 wild Red Junglefowls (RJF) and 27 domesticated chickens, including 10 Tibetan chickens (TBC) from Tibet, 8 village chickens (VC) from the lowland plains of Yunnan adjacent to Tibet, 8 Xishuangbanna game fowl (XGF) that are used for cockfighting in southeastern Asia, and 1 individual from a commercial chicken breed, with an average 16.3 fold coverage (**Supplementary Table 16**). The phylogenetic relationship among these individuals was constructed using the weighted neighboring joining method (Bruno et al. 2000), and the clustering on the resulting

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tree showed two distinct groups of Tibetan chickens (**Fig. 1A**). Similarly, two separated groups of Red Junglefowls were found in the phylogeny. This phylogenetic pattern was also supported by a multidimensional scaling analysis of the matrix of genome-wide IBS (identity-by-state) pairwise distances (**Supplementary Fig. 12**), by principal component analysis (**Fig. 1B**), and Bayesian model-based analyses (**Fig. 1C**). These results collectively indicated that domestic chickens can be traced back to more than one single genetic source, suggesting multiple origins or origin events for domesticated chickens, a conclusion that was previously proposed by earlier analyses of mtDNA sequences (Liu et al. 2006; Miao et al. 2012). Previous comparison of runs of homozygosity (ROH), which represent long stretches of consecutive homozygous genotypes, revealed higher ROH values and lower heterozygosity among commercial chickens, Silkies, and the Taiwanese native chicken L2 (Fan et al. 2013), which is likely attributable to inbreeding or population founder effects (**Fig. 1D**). From our current samples, group I Tibetan chickens showed the highest amounts of linkage disequilibrium (LD), suggesting high inbreeding potentially due to artificial selection **(Fig. 1E**).

Our phylogenetic analyses showed two distinct groups of Tibetan chickens, but it was not clear whether one of these groups of breeds arose from a crossing between the other group of Tibetan chickens and other breeds of chicken. Using TreeMix (Pickrell and Pritchard 2012) to infer migration patterns among the different breeds, we found a genetic contribution from common commercial chickens into group II of the Tibetan chicken (**Fig. 2**), which is consistent with recent breeding activities in

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Tibet, where commercial breeds have been introduced (Wu and Li 2012), but this analysis did show any clear gene flow between the two groups of Tibetan chickens (**Fig. 2**). Further admixture analyses, including the *f3* test (Reich et al. 2009) (**Supplementary Table 17**) and the D-statistics (ABBA/BABA test) (Green et al. 2010) (**Supplementary Table 18**), also showed no evidence for gene flow between the two groups of Tibetan chickens, but similar to the TreeMix finding, did show gene flow between commercial chickens and group II of the Tibetan chicken. These findings suggest that the multiple origins for the Tibetan chickens likely can be traced to two independent sources. Further studies with more populations from Asia, especially from surrounding regions are necessary to resolve the origin of the Tibetan chickens.

Population genomic analyses revealed demographic history of chickens

Using a pairwise sequential Markovian coalescent (PSMC) model (Li and Durbin 2011) to infer the demographic history of the ancestors of the chickens using individuals with high genomes coverage (>20X) (**Supplementary Fig. 13**) showed that the population size of the chicken ancestor increased sometime around 1 Myr ago, reaching a peak at ~100kyr years ago, with a dramatic decrease following that peak, which matches the known environmental changes in Southeast Asia where the Red Junglefowl are distributed. During the Quaternary ice ages (~2.4 Myr ago to present), the rainforests of Southeast Asia, which is one of the world's biodiversity hotspots containing a high concentration of endemic species (Sodhi et al. 2004), served as a

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refugia, enabling the persistence of forest biota (Gathorne-Hardy et al. 2002; Sodhi et al. 2004). However, a substantial contraction of the forest area occurred in north equatorial Southeast Asia region during the Last Glacial Period (~125–10 kyr ago) (Wurster et al. 2010), which greatly reduced biodiversity and population sizes of species residing in this area.

Potential independent signatures of selection in the two Tibetan chicken populations

Regions under directional selection should show specific signatures of variation, including high population differentiation, lower levels of nucleotide diversity and long range haplotype homozygosity (Sabeti et al. 2006). To determine whether direction selection might have occurred in the Tibetan chicken, we first explored the genomic landscape of the population differentiation to identify candidate genes that potentially control specific adaptive traits present in the two groups of Tibetan chickens. Population differentiation from all other chickens was evaluated for each group of Tibetan chickens for each relevant SNP. Within the two regions showing extremely high F_{ST} values (top 0.01%) in group I, only one protein coding gene, *SLC35F1* located on chromosome 3, was identified (**Fig. 3A**). No non-synonymous SNPs were found within the region, suggesting that the potential target for selection were regulatory mutations. While the function of *SLC35F1* is not known, genome-wide association studies (GWAS) have found an association between variants of *SLC35F1* and the resting heart rate in humans (Eijgelsheim et al. 2010).

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A total of 203 candidate genes were identified using a sliding window analysis of the group I lineage for genes that showed F_{ST} values that were in the top 1%, and thus were considered to be potentially positively selected (**Fig. 3A**). The windows retained statistical significance when compared with simulated data generated using a model of the potential demographic history of these chickens (**Supplementary Fig. 14**). Analysis of gene enrichment within this set of genes found significant enrichment for genes potentially involved in high-altitude adaptation (**Table 2**, **Supplementary Table 19**). Gene enrichment analysis can yield high false-positive rates (Pavlidis et al. 2012), thus additional functional and physiological experiments are needed to verify the contributions of these genes. Among the candidate genes, four (*AATF*, *CYBB*, *FOXM1*, and *BCDO2*) are involved in the "reactive oxygen species metabolic process" (GO: 0072593) ($P<0.05$) and another 4 (*ARVCF*, *COMT*, *SDHD*, *COL10A1*) are enriched in the Human Phenotype Category "abnormality of calcium homeostasis" (P= 6.22E-03). Hypoxia should cause a significant increase in the concentration of intracellular reactive oxygen species (ROS) in pulmonary cells (Shimoda and Undem 2010; Wang and Zheng 2010), and accordingly ROS should then lead to large increase in intracellular concentration of Ca^{2+} mediated by multiple ion channels (Shimoda and Undem 2010; Wang and Zheng 2010). Both ROS and Ca^{2+} are important signaling molecules that regulate the response to hypoxia. Ca^{2+} modulates cell contraction, cell proliferation and growth (Shimoda and Undem 2010; Wang and Zheng 2010), and calcium signaling will stimulate the translation of HIF-alpha, an ubiquitous transcription factors that mediates adaptation to hypoxia (Hui et al. 2006) (**Fig. 4**).

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In the group II lineage, only one protein-coding gene, *RYR2* (ryanodine receptor 2), was found adjacent to a region with extremely high F_{ST} values (top 0.01%) (**Fig. 3B**), suggesting that the potential target of selection was likely a regulatory sequence for *RYR2*. Hypoxia typically causes a release of intracellular Ca^{2+}, mediated by the ryanodine receptors, which then leads to increased cell contraction (Wang and Zheng 2010). RYR2 functions as a component of calcium channels that supply Ca^{2+} for cardiac muscle excitation-contraction coupling. Mutations in this gene are associated with many ventricular diseases (e.g. heart failure) (Marks 2013). A sliding window analyses yielded 176 candidate genes located in the top 1% regions with high F_{ST} value that may also be potential targets of selection in the group II lineage (**Fig. 3B**). These regions also demonstrate statistical significance when compared with simulated data modeled from the potential demographic history (**Supplementary Fig. 15**). These candidate genes are enriched for many Human Phenotype categories associated with diseases of the heart and lung, such as obstructive lung disease, chronic obstructive pulmonary disease, respiratory difficulties, emphysema, hypertension, hypotension, abnormality of cardiovascular system physiology, and dyspnea (**Supplementary Table 20**), thus some of these genes may contribute to adaptive development of the heart and lung allowing the Tibetan chickens to survive in their high-altitude environment. An earlier report had found some specific histological adaptations present in the Tibetan chicken, but not their lowland counterparts, such as increased heart and lung size (Wu and Li 2012).

Only 7 genes were found to overlap between the groups of candidate genes

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identified for the two Tibetan chicken lineages, and their potential functional association with high altitude adaptation of these genes was unclear (**Supplementary Table 21**). In addition, the level of population differentiation of the Tibetan chicken group I from other chickens was calculated, as was the population differentiation of Tibetan chicken group II from other chickens, for each gene. We retrieved genes displaying the top 1% population differentiation from other chickens for both Tibetan chicken groups, but only 5 protein coding genes overlapped between these two gene sets (**Supplementary Fig. 16**). The distribution of the highly differentiated SNPs (F_{ST}>0.6) in the exons, introns, UTRs, and intergenic regions of the genes was examined. Tibetan chickens in group I harbor a significantly greater number of highly differentiated non-synonymous SNPs than UTR SNPs. However, for Tibetan chicken group II, the inverse pattern was observed, with significantly more highly differentiated UTR SNPs than non-synonymous SNPs (**Supplementary Fig. 17**). Overall, this data suggests that independent molecular mechanisms of high altitude adaptation occurred on these two different lineages, although additional functional and physiological experiments are needed to verify this hypothesis.

Constrained non-coding elements (CNEs) are abundant in vertebrate genomes, with most serving as important gene regulatory modules that drive gene expression, and can potentially be involved in adaptive evolution (Kim and Pritchard 2005; Woolfe et al. 2005; Lowe et al. 2011). We retrieved CNEs containing highly differentiated SNPs (F_{ST}>0.6) in Tibetan chicken groups I and II (see supplementary material), and identified candidate enriched categories significantly related to lung,

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vein and heart development, and respiratory related categories, such as '*abnormal anterior cardinal vein morphology*', '*absent myocardial trabeculae*' and '*respiratory failure*' in the Tibetan chicken group I genes (**Supplementary Dataset 6**); and '*respiratory distress*', '*abnormal breathing pattern*', '*respiratory system development*' in the Tibetan chicken group II genes (**Supplementary Dataset 6**). These genes might contribute to the adaptive evolution of the development of the heart and respiratory systems of Tibetan chickens living at high altitude, although the exact functions of these CNEs are largely unclear.

Genomic regions experiencing directional selection should show the specific signature of lower levels of nucleotide diversity and long range haplotype homozygosity (Sabeti et al. 2006). To better understand selection on the group II lineage, which has a relatively larger sample size than group I, we used the signature of a lower level of nucleotide diversity to identify regions that might have experienced selective sweeps. In total, 363 candidate genes were identified within regions with significant lower levels of nucleotide diversity (π) and low Tajima's D values. Genes representing the "calcium signaling pathway" were overrepresented in this gene set, with 8 identified genes (**Supplementary Table 22**), paralleling the significant enrichment of candidate selected genes from group I with an important role for Ca^{2+} in the response to hypoxia. Likewise, 2 "citrate cycle (TCA cycle)" pathway genes (*SDHA*: succinate dehydrogenase; *FH*: fumarate hydratase, mitochondrial) were over-represented, which may represent an adaptation of energy metabolism to high altitude.

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In general, variants experiencing directional selection usually show both strong long range haplotype homozygosity and linkage disequilibrium as the frequencies of the variants rapidly increase, as recombination will not have substantially broken the haplotypes (Sabeti et al. 2002; Sabeti et al. 2006). To explore this feature of the selective landscape in group II Tibetan chickens, we employed the XP-EHH method (Sabeti et al. 2007) to identify potential variants experiencing selection. A sliding window analysis was performed, resulting in the identification of 193 genes. Gene enrichment analysis found that a group of 4 of these genes belong to the significantly over-represented category "calcium signaling pathway" (**Supplementary Table 23**), a result that was also supported using a 20kb sliding window analysis (**Supplementary Table 24**).

XP-EHH analysis also showed that the four SNPs with highest XP-EHH values are located within one gene, *HDGFRP3* (hepatoma-derived growth factor, related protein 3). Knockdown of *HDGFRP3* in A549 cells (adenocarcinomic human alveolar basal epithelial cells) was previously shown to promote apoptosis and to be associated with the generation of substantial quantities of reactive oxygen species (ROS) (Yun et al. 2013), a response similar to hypoxia. We further examined 100 SNPs with the highest XP-EHH values, which map to 10 regions shorter than 20kb and overlap with 8 genes (**Fig. 3C**). Among these 8 genes, 2 are involved in the Ca^{2+} signaling pathway. *CACNA1G* (calcium channel, voltage-dependent, T type, alpha 1G subunit) encodes one of the voltage-sensitive calcium channels mediating the entry of calcium into cells, and *ANO1* (or *Tmem16a*), is an intrinsic constituent of the calcium-dependent chloride

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channel (Caputo et al. 2008), the activity of which is associated with pulmonary hypertension (Forrest et al. 2012).

From our board spectrum of analyses 6 and 16 genes were found to overlap genes within regions showing signatures of higher F_{ST} and XP-EHH values, and lower levels of nucleotide diversity and Tajima's D values in the group II and I lineages of Tibetan chickens, respectively (**Supplementary Figs. 18-19**). Among the genes identified in these two lineage, the gene *PDGFRA* (platelet-derived growth factor receptor, alpha polypeptide) was found to be candidate gene undergoing potentially selection on the group II lineage, while *PDGFC*, which encodes one of ligands for the platelet-derived growth factor receptor, was a candidate selected gene in the group I lineage. PDGF-mediated activation of PDGFR was previously shown to result in the activation of NADPH oxidases, which increase the levels of ROS (Lei and Kazlauskas 2014). Likewise, PDGFRA could be deregulated by HIF1-alpha and HIF2-alpha (Hanna et al. 2013).

On the whole, our integrative analysis suggests that the calcium signaling pathway was an important target of selection for hypoxic high-altitude adaptation in Tibetan chickens as they accompanied their human domesticators into the highlands of the Tibetan Plateau (**Fig. 4**). No evidence of directional selection for genes in this pathway in any other animal that inhabits high-altitude areas has been reported to date. As such, the candidate selected genes identified here, and their variants, may be useful targets for clarifying our understanding of the domestication of chickens in Tibet, as well as in current chicken breeding efforts to develop new breeds for the highlands.

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This latter point is especially significant since eggs from Tibetan chickens have a hatchability of >70% in their high-altitude environment, while eggs from lowland chicken breeds have less than 40% at high altitude (Wu and Li 2012).. This difference in hatchability is a significant factor restricting the breeding other breeds of chicken in Tibet (Wu and Li 2012). Similarly, our findings may also have some clinical applications, as tumor hypoxia has gained increasing attention in the development of cancer therapy (Wilson and Hay 2011).

Materials and Methods

De novo genome sequencing, assembly and annotation

A female Tibetan chicken (from Nyingchi in Tibet, China) was used for de novo genome sequencing. 9 pair-end libraries with insert sizes of 250bp, 500bp, 800bp, 2kb, 5kb, 10kb and 20kb were constructed, and sequenced on the Illumina Genome Analyzer platform. Reads, after quality filtration, were assembled by SOAPdenovo with the processes of contig construction, gap filling and scaffold construction (Li et al. 2010). Full description of the de novo genome assembly is in Supporting Information S1. Gene sets were predicted by integrating homology-based, ab initio derived, and RNA-seq based methods. Gene functions were assigned according to the best match of the alignments by searching protein, protein domain, GO, and KEGG databases.

Full description of the gene annotation is in Supporting Information S2.

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Population genome re-sequencing, read mapping and SNP calling

Whole genomes of 32 chickens (5 wild Red Junglefowls, 10 Tibetan chickens, 8 village chickens, 8 Xishuangbanna game fowl, and 1 commercial breed Roman laying hen) were re-sequenced on an Illumina Hiseq 2000 Sequencer. Genome sequencing data from another Tibetan chicken generated by Illumina for assembling the de novo genome of the Tibetan chicken was also used for the analysis. In addition, previously published genome sequence data from 2 other chicken, 1 Silkie and 1 Taiwanese native chicken (Fan et al. 2013) with sample accession number SAMN02142123 and SAMN02142124 and downloaded from http://www.ebi.ac.uk, were used in this analysis.

Pair-end reads after quality filtration were mapped onto the chicken reference genome using BWA-MEM (bio-bwa.sourceforge.net). Bam files were sorted using SortSam and duplicated reads was marked using MarkDuplicates from Picard-Tools 1.56 (http://picard.sourceforge.net). Reads that mapped around the edges of indels with mismatches were realigned and base quality scores that estimated with biases were recalibrated using IndelRealigner and BaseRecalibrator command from Genome Analysis Toolkit 2.6 (www.broadinstitute.org/gatk). SNPs were detected and filtered using UnifiedGenotyper and VariantFiltration command in Genome Analysis Toolkit 2.6.

Analysis of the demographic history, population variation, and admixture

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Population demographic history was predicted by PSMC (Li and Durbin 2011) with generation time (g=1 year) and mutation rates per year ($u=1.91*10^{-9}$) (Nam et al. 2010) using single individual genomes. The phylogenetic relationships of the individual genome sequences were estimated buy a tree constructed by the weighted neighbor joining method using pair-wide distances among the chickens. Population structure was deduced with the program STRUCTURE (Pritchard et al. 2000). Runs of homozygosity (ROH) and linkage disequilibrium (LD) were calculated using PLINK (Purcell et al. 2007).

Migration events among the populations were first inferred by TreeMix (Pickrell and Pritchard 2012) allowing two, three, and four migration events. According to Fig. 1A, two RJF samples were assigned as the "outgroup" to root the tree. The covariance matrix was estimated based on 1,314,904 unlinked sites with a window size of 10. Additional admixture analyses, including the f3 test (Reich et al. 2009) and the D-statistics (ABBA/BABA test) (Green et al. 2010) were performed using TreeMix (Pickrell and Pritchard 2012) and ADMIXTOOLS (Patterson et al. 2012), respectively.

Identification of candidate genes under positive selection in the Tibetan chicken lineages

For each group of Tibetan chickens (I and II), population differentiation (F_{ST}) from all other chickens was calculated for each SNP as described in (Akey et al. 2002).

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Negative values have no biological explanation and were arbitrarily set to 0. For the analysis of the extended haplotype homozygosity and haplotypes of each chromosome were deduced using shapeit.v2.r727 (Delaneau et al. 2013) (http://www.shapeit.fr/). XP-EHH value for each SNP was calculated using the XPEHH program (http://hgdp.uchicago.edu/software/). Sliding window analyses were performed with a window size of 100kb and a step size of 50kb. The average F_{ST} and XP-EHH values of SNPs in each window were calculated. Here, we used the outlier method based on the genome wide empirical distribution of these statistics for detecting genomic regions potentially targeted by positive selection. Windows with the top 1% F_{ST} or XP-EHH values were retrieved with the Variant Effect Predictor online tool (http://www.ensembl.org). Protein coding genes in these outlier windows were treated as candidate positively selected genes. For the analysis of the signature of lower nucleotide diversity, nucleotide diversity (π) and Tajima's D values of each window were calculated for the Tibetan chickens (I and II), and the population of all other chickens. Protein coding genes at the regions with higher values of ($\pi_{(others)} - \pi_{(Tibetan\ chickens)}$), and Tajima's $D_{(others)}$ - Tajima's $D_{(Tibetan\ chickens)}$ (top 5%) were retrieved as candidate positively selected genes. Gene enrichment analysis were performed with g:profiler (http://biit.cs.ut.ee/gprofiler/index.cgi). Benjamini-Hochberg FDR (False discovery rate) was used for correcting the P-values. The Gene Ontology categories "Molecular Function" (MF), "Biological Process" (BP), and "Cellular Component" (CC), and Human Phenotype (HP) categories were used in these analyses.

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Figure legends

Figure 1: Analysis of population variation and structure of the Tibetan chickens and their relatives. (A) Phylogenetic relationship of the Tibetan chickens and relatives. (B) PCA (Principal component analysis). (C) Population structure; K=2 is the best. (D) ROH. (E) Linkage disequilibrium (LD) decay. RJF: Red Junglefowls; TBC: Tibetan chickens; VC: village chickens; XGF: Xishuangbanna game fowl; C: commercial chickens: S: Silkie; L: Taiwanese native chicken L2. Different colors (B, E) represent different breeds, as in (A).

Figure 2: Inferred chicken tree of mixture events deduced by TreeMix. Abbreviations are the same as in Figure 1A. Two RJF samples were assigned as "outgroup" to root the tree. The graph inferred for the chicken populations, allowing two (A), three (B), and four (C) migration events. Migration arrows are colored according to their weight. Horizontal branch lengths are proportional to the amount of genetic drift that has occurred on each branch. The scale bar shows ten times the average standard error of the entries in the sample covariance matrix.

Figure 3: Signature of selection among the Tibetan chickens. Genomic landscape of the population differentiation in groups I (A) and II (B) of Tibetan chickens. F_{ST} values around genes *SLC35F1* and *RYR2* within the regions showing extremely high F_{ST} values (top 0.01%) in group I (A) and group II (B) are

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plotted. (C) Genomic landscape of the XP-EHH values in group II Tibetan chickens using single SNPs.

Figure 4: Genes involved in the Ca^{2+} signal pathway. Genes in red are putatively under selection in the group II lineage. Hypoxia would induce the release of ROS (reactive oxygen species) and cause extracellular Ca^{2+} influx mediated by ion channels and intracellular Ca^{2+} release by ryanodine receptors (RYRs), which can also be induced by ROS. Ca^{2+} will then cause cell contraction, an important response to hypoxia.

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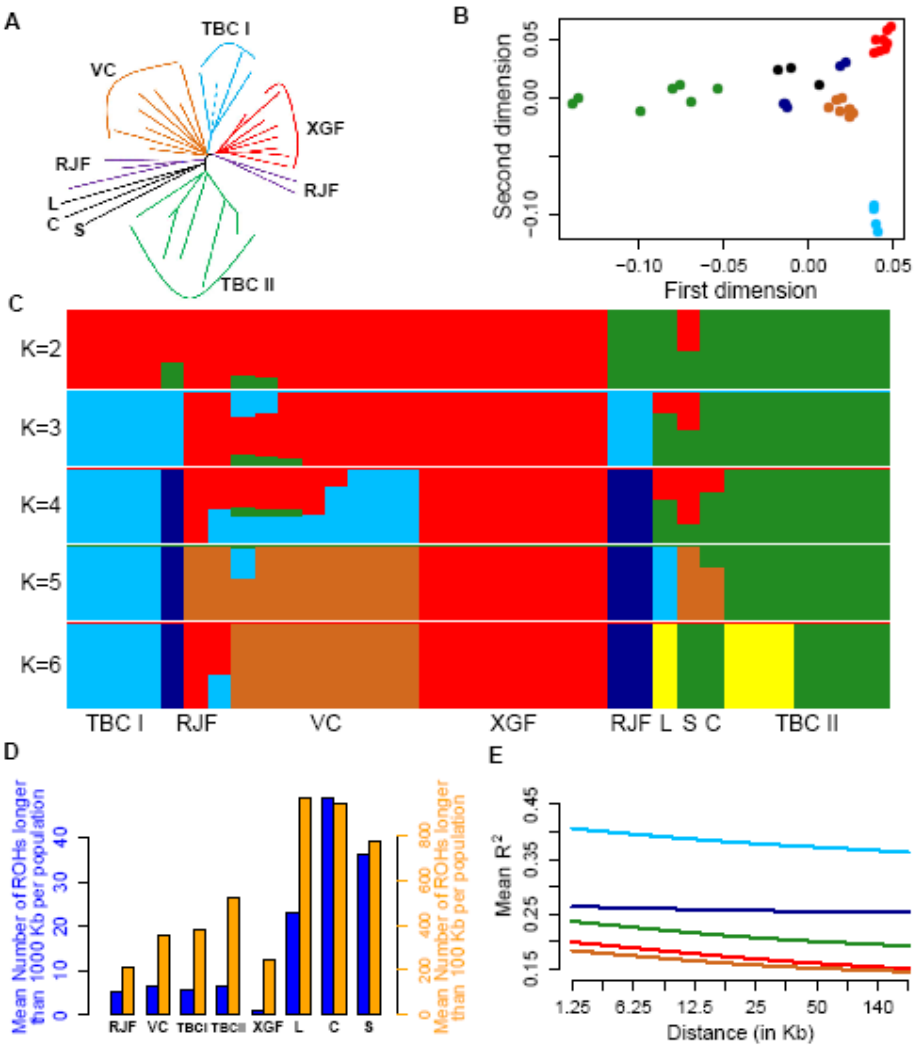
Table 1: Gene enrichment analysis of genes showing high population differentiation between group I Tibetan chickens and other chickens.

ID	Type	Description	N	P-value
GO:0021575	BP	hindbrain morphogenesis	3	2.82E-02
GO:2000377	BP	regulation of reactive oxygen species metabolic process	3	4.35E-02
GO:0072593	BP	reactive oxygen species metabolic process	4	4.97E-02
HP:0010929	hp	abnormality of cation homeostasis	6	6.22E-03
HP:0003326	hp	myalgia	4	7.63E-03
HP:0001063	hp	acrocyanosis	3	9.18E-03
HP:0004363	hp	abnormality of calcium homeostasis	4	1.43E-02
HP:0003111	hp	abnormality of ion homeostasis	6	1.44E-02
HP:0010927	hp	abnormality of divalent inorganic cation homeostasis	4	1.64E-02
HP:0001608	hp	abnormality of the voice	6	1.91E-02
HP:0002721	hp	immunodeficiency	4	2.00E-02
HP:0000964	hp	eczema	4	2.26E-02
HP:0100851	hp	abnormal emotion/affect behavior	8	2.29E-02
HP:0001601	hp	laryngomalacia	3	2.46E-02
HP:0011123	hp	inflammatory abnormality of the skin	6	2.48E-02
HP:0005374	hp	cellular immunodeficiency	3	2.69E-02
HP:0002515	hp	waddling gait	3	3.18E-02
HP:0001005	hp	dermatological manifestations of systemic disorders	5	3.26E-02
HP:0003457	hp	EMG abnormality	5	3.68E-02
HP:0000961	hp	cyanosis	3	3.73E-02
HP:0001437	hp	abnormality of the musculature of the lower limbs	3	4.64E-02
HP:0001939	hp	abnormality of metabolism/homeostasis	22	5.00E-02
KEGG:00510	ke	N-Glycan biosynthesis	4	5.00E-02
REAC:456716	re	Purine biosynthesis	3	5.00E-02

Note: Only pathways with more than 2 genes are shown. N is the number of genes. P-values are corrected by Benjamini-Hochberg FDR.

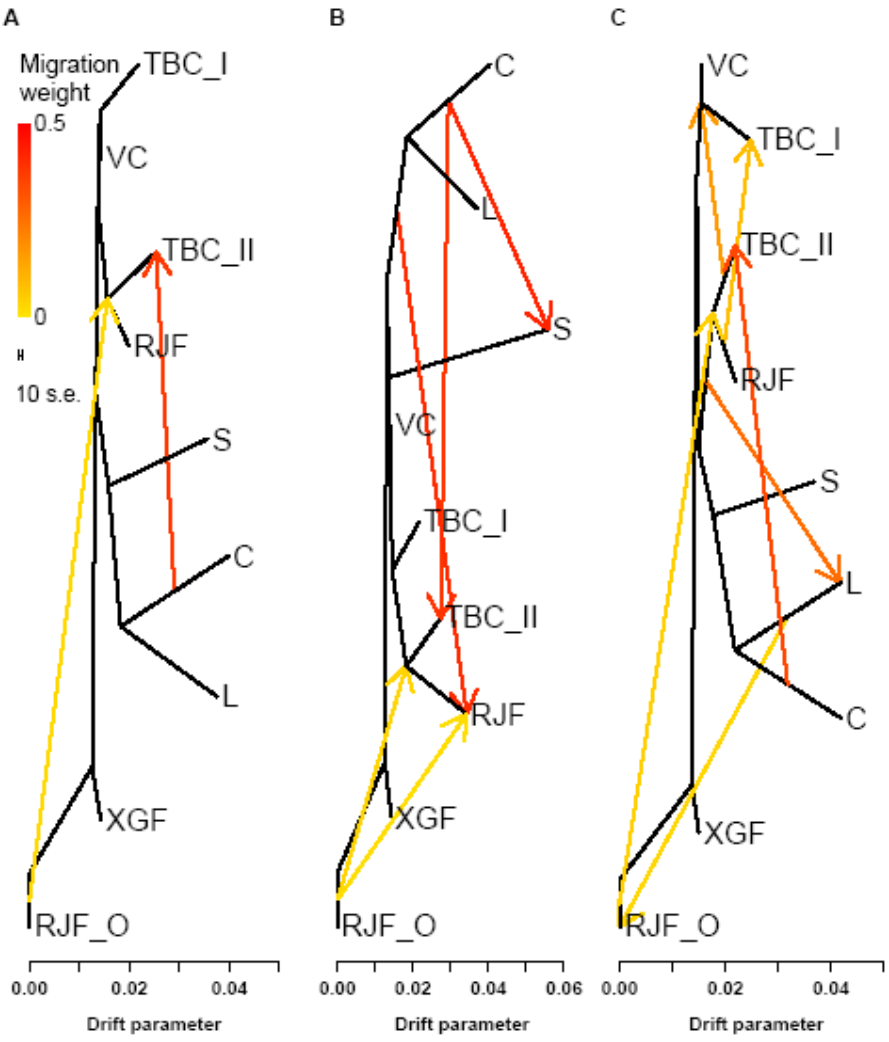
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Figure 1



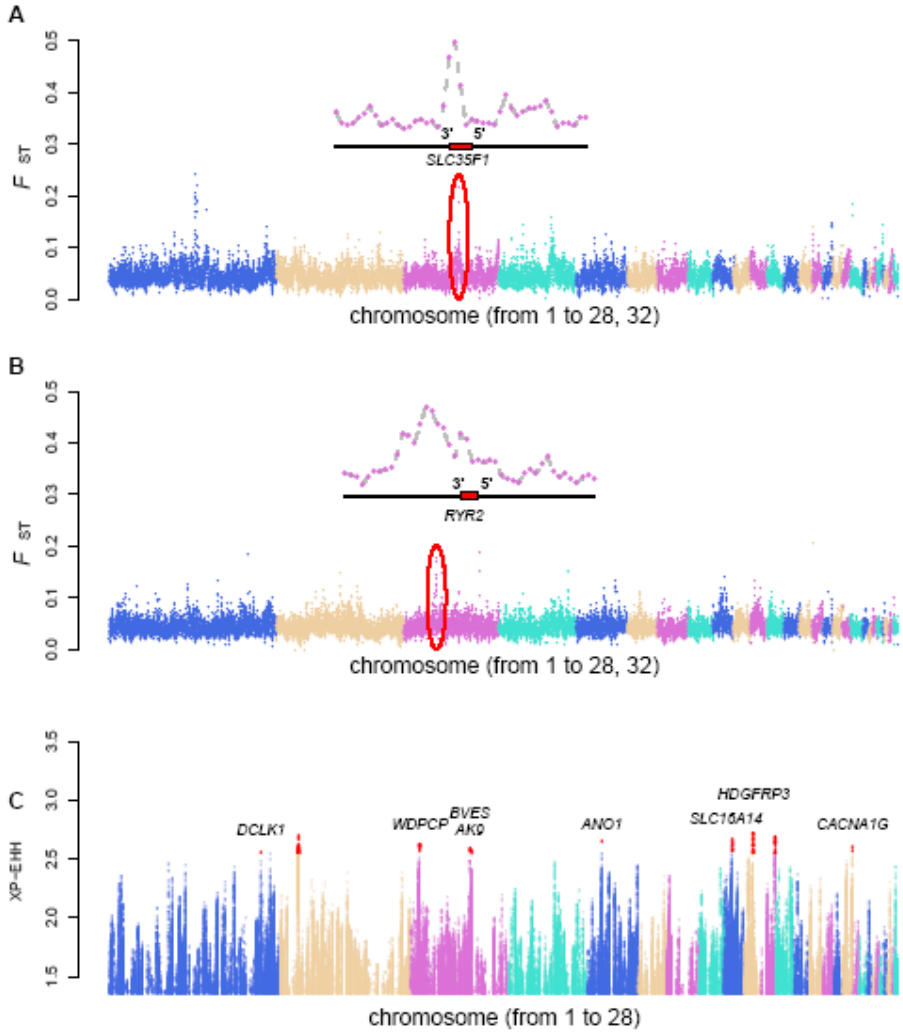
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Figure 2



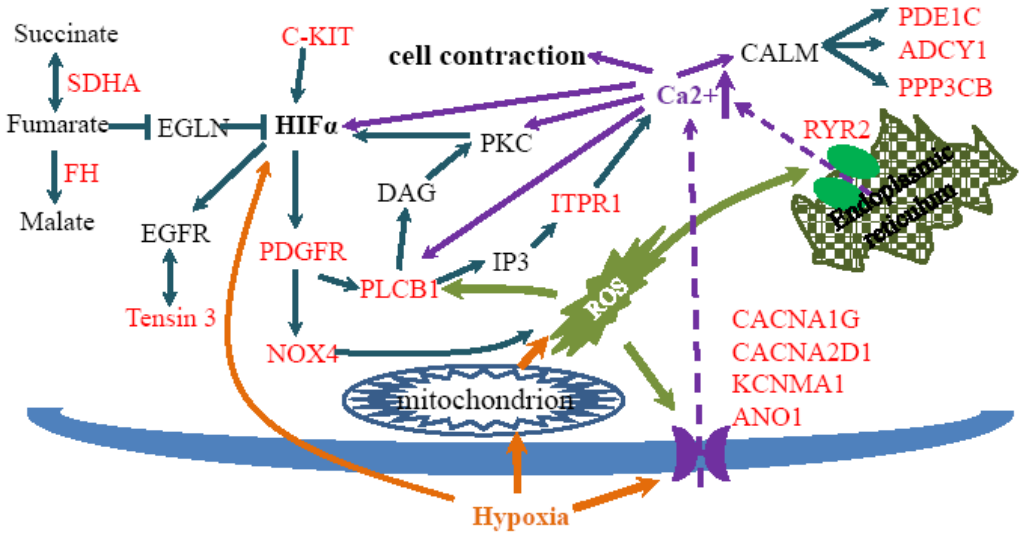
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Figure 3



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Figure 4



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